CHAPTER 10

THERMAL PROPERTIES OF MATTER

CHAPTER ESSENCE AT A GLANCE:
Heat is energy in transit! When materials heat up, atoms jiggle farther apart causing thermal expansion. But water is nature's miracle: between 0°C and 4°C, water shrinks instead of expanding! That's why ice floats and fish survive frozen lakes!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Thermal Expansion:** $\Delta L = \alpha L \Delta T$ (Linear); $\Delta A = 2\alpha A \Delta T$ (Superficial); $\Delta V = 3\alpha V \Delta T$ (Volume). Relation between coefficients: $\alpha : \beta : \gamma = 1 : 2 : 3$.

• **Anomalous Expansion of Water:** Water contracts from 0°C to 4°C, reaching MAXIMUM density at 4°C (1000 kg/m^3). In winter, 4°C water sinks to the bottom, allowing fish to survive under surface ice!

• **Latent Heat vs Sensible Heat:** Sensible heat causes temperature change: $Q = m s \Delta T$. Latent heat causes phase change at CONSTANT temperature: $Q = m L$. Steam at 100°C causes worse burns than boiling water due to extra latent heat of vaporization (2.26×10^6 J/kg)!

• **Heat Transfer Modes:** Conduction (molecule-to-molecule vibration in solids), Convection (actual bulk fluid movement), Radiation (electromagnetic waves, fastest, needs no medium).

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\Delta L = \alpha L \Delta T, \quad \Delta A = 2\alpha A \Delta T, \quad \Delta V = 3\alpha V \Delta T$$

Linear, Areal, and Volumetric Thermal Expansion

$$Q = mc\Delta T, \quad Q = mL$$

Sensible Heat and Latent Heat of Phase Change

$$\frac{dQ}{dt} = -KA\frac{dT}{dx}$$

Fourier's Law of Thermal Conduction

$$\frac{dT}{dt} = -k(T - T_0)$$

Newton's Law of Cooling

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Latent heat occurs with ZERO change in temperature! During melting (ice to water) or boiling (water to steam), all heat goes into breaking molecular bonds.

• **Alert:** Density of water is MAXIMUM at 4°C (1.000 g/cm^3). Remember this for 1-mark conceptual board questions!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Explain anomalous expansion of water and state its ecological significance for aquatic life during winter. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Anomalous Expansion]** Unlike most liquids that contract on cooling, water expands when cooled from 4 °C down to 0 °C. Water attains its maximum density at 4 °C. [1 Mark]

• **[Step 2: Ecological Significance]** In freezing winters, surface water cools to 4 °C, becomes dense, and sinks to the bottom. Once the entire lake reaches 4 °C, further cooling forms ice (density < water) at the top surface (0 °C), which acts as an insulator. The lake bottom stays at 4 °C, enabling fish and aquatic life to survive beneath the frozen surface. [1 Mark]

• **Key Exam Result:** Water has max density at 4 °C; protects aquatic life under frozen ice

2. Define specific heat capacity (s) and molar heat capacity (C). State their SI units. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Specific Heat Capacity]** The amount of heat required to raise the temperature of unit mass (1 kg) of a substance by 1 Kelvin: s = ΔQ divided by (m ΔT) (SI unit: J kg^{-1} K^{-1}). [1 Mark]

• **[Step 2: Molar Heat Capacity]** The heat required to raise temperature of 1 mole of a substance by 1 Kelvin: C = ΔQ divided by (μ ΔT) (SI unit: J mol^{-1} K^{-1}). [1 Mark]

• **Key Exam Result:** s in J kg^{-1} K^{-1}, C in J mol^{-1} K^{-1}

3. Define coefficients of thermal expansion: linear (α), superficial (β), and volume (γ). State the relation between them. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Definitions]** 1. Linear expansion: α = ΔL divided by (L ΔT). 2. Superficial (area) expansion: β = ΔA divided by (A ΔT). 3. Volume (cubical) expansion: γ = ΔV divided by (V ΔT). [1.5 Marks]

• **[Step 2: Inter-relation Proof]** For an isotropic cube of side L: A = L^2 => dA divided by dT = 2 L (dL divided by dT) => β = 2 α. V = L^3 => dV divided by dT = 3 L^2 (dL divided by dT) => γ = 3 α. Therefore: α : β : γ = 1 : 2 : 3 (or β = 2 α, γ = 3 α). [1.5 Marks]

• **Key Exam Result:** α : β : γ = 1 : 2 : 3

4. Define thermal conductivity (K). Derive an expression for the rate of heat flow through a metallic slab under steady state. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Factors Influencing Heat Flow]** Consider a slab of face area A and thickness L with faces maintained at temperatures T1 and T2 (T1 > T2). Under steady state, heat Q conducted in time t is: 1. proportional to area A. 2. proportional to temperature difference (T1 - T2). 3. proportional to time t. 4. Inversely proportional to thickness L. [1.5 Marks]

• **[Step 2: Formula & Coefficient K]** Q ∝ [A (T1 - T2) t] divided by L ⇒ Q = K A (T1 - T2) t divided by L. Rate of heat flow H = dQ divided by dt = K A (T1 - T2) divided by L. K is thermal conductivity of material (SI unit: W m^{-1} K^{-1}). [1.5 Marks]

• **Key Exam Result:** Rate of Heat Flow H = K A (T1 - T2) divided by L

5. State Newton's Law of Cooling. Derive the mathematical expression for the cooling of a body. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Statement]** The rate of loss of heat of a body by radiation is directly proportional to the difference in temperature between the body and its surroundings, provided the temperature difference is small: -dQ divided by dt = k (T - T 0). [1 Mark]

• **[Step 2: Temperature Form]** Since Q = m s T: dQ divided by dt = m s (dT divided by dt). -m s (dT divided by dt) = k (T - T 0) => dT divided by dt = -K (T - T 0), where K = k divided by (m s). [1 Mark]

• **[Step 3: Integration]** dT divided by (T - T 0) = -K dt. Integrating: ln(T - T 0) = -K t + C ⇒ T(t) = T 0 + C e^(-K t). [1 Mark]

• **Key Exam Result:** -dT divided by dt = K (T - T 0) => ln(T - T 0) = -K t + C

6. A metal ball of mass 0.1 kg is heated to 100 °C and placed in a copper calorimeter containing 150 g of water at 20 °C. The final steady temperature is 25 °C. If water equivalent of calorimeter is 10 g, find the specific heat of the metal. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Principle of Calorimetry]** Heat lost by hot metal ball = Heat gained by water + calorimeter. [0.5 Mark]

• **[Step 2: Heat Lost by Metal]** Mass of metal m m = 0.1 kg = 100 g. Temperature fall Δ T m = 100 - 25 = 75 °C. Heat lost Q lost = m m s m Δ T m = 100 x s m x 75 = 7500 s m cal. [1 Mark]

• **[Step 3: Heat Gained by Water and Calorimeter]** Mass of water m w = 150 g, water equivalent w = 10 g ⇒ effective mass M = 150 + 10 = 160 g. Temperature rise Δ T w = 25 - 20 = 5 °C. Specific heat of water s w = 1 cal g^{-1} $°C^{-1}$. Heat gained Q gain = M s w Δ T w = 160 × 1 × 5 = 800 cal. [1 Mark]

• **[Step 4: Solve for s m]** 7500 s m = 800 ⇒ s m = 800 divided by 7500 = 0.1067 cal g^{-1} $°C^{-1}$ = 0.1067 × 4186 J kg^{-1} K^{-1} ≈ 446.6 J kg^{-1} K^{-1}. [0.5 Mark]

• **Key Exam Result:** Specific heat of metal = 446.6 J kg^{-1} K^{-1}

7. A pan filled with hot food cools from 94 °C to 86 °C in 2 minutes when the room temperature is at 20 °C. How long will it take to cool from 71 °C to 69 °C? [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Newton's Law of Cooling Average Form]** (T1 - T2) divided by t = K [(T1 + T2) divided by 2 - T 0]. [1 Mark]

• **[Step 2: First Cooling Phase (94 to 86 in 2 min)]** (94 - 86) divided by 2 = K [(94 + 86) divided by 2 - 20] ⇒ 8 divided by 2 = K [90 - 20] ⇒ 4 = 70 K ⇒ K = 4 divided by 70 = two-thirds5 min^{-1}. [1.5 Marks]

• **[Step 3: Second Cooling Phase (71 to 69)]** (71 - 69) divided by t = K [(71 + 69) divided by 2 - 20] => 2 divided by t = K [70 - 20] = 50 K. Substituting K = two-thirds5: 2 divided by t = 50 x (two-thirds5) = 100 divided by 35 => t = 2 × 35 divided by 100 = 70 divided by 100 = 0.7 minutes = 42 seconds. [1.5 Marks]

• **Key Exam Result:** Time taken = 0.7 minutes (42 seconds)